OFFICIAL USER GUIDE

ADSI Inverter Dashboard

Professional plant monitoring, energy analytics, forecast generation, and controlled remote coordination for utility-scale solar inverter fleets.

Version 2.8.1

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March 2026

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Overview & Architecture

ADSI Inverter Dashboard provides centralized operational visibility for a utility-scale solar inverter fleet. The application consolidates live status, interval energy accumulation, alarms, audit activity, day-ahead generation forecasting, and formal reporting into one workstation interface built on Electron.

997.64 KW PER INVERTER (PMAX)	**4** NODES / INVERTER	**249.41** KW PER NODE (PMAX)
906.92 KW DEPENDABLE (PNOM)		

Live Monitoring

Real-time inverter and node-level telemetry with supervised START/STOP control actions.

Analytics & Forecasting

Interval charts with ML-trained day-ahead forecast comparison and Solcast tri-band integration.

Alarm Management

Episode tracking with severity, duration, bitmask change logic, and quick-ACK notifications.

Reporting & Export

Daily performance reports with availability calculation and multi-format structured exports.

Remote Operations

Gateway-linked remote viewing and write control via Tailscale, VPN, or direct LAN.

Cloud Backup

Scheduled backup with OneDrive, Google Drive, and S3-compatible storage integration.

System Architecture

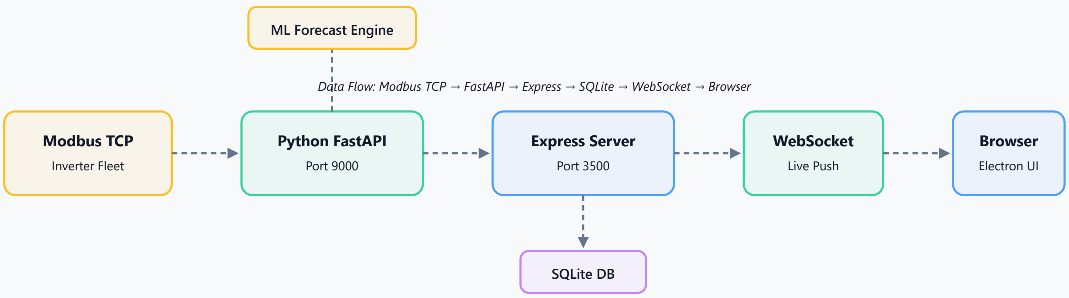

Fig 1 — ADSI Dashboard system architecture and data flow pipeline

Technology Stack

LAYER	TECHNOLOGY	ROLE
Desktop Shell	Electron 29	Native window management, file dialogs, licensing
API Server	Express 4	REST API, polling orchestration, WebSocket broadcast
Database	SQLite via `better-sqlite3`	Persistent storage for telemetry, reports, forecasts
Frontend	Vanilla JS + Chart.js 4	Real-time UI rendering and interactive charts
Inverter Service	Python FastAPI (port 9000)	Modbus TCP polling and inverter command dispatch
Forecast Engine	Python ML (LightGBM)	Day-ahead generation model with Solcast tri-band

System Requirements

REQUIREMENT	DETAILS
Operating System	Windows 10 / 11 (64-bit)

REQUIREMENT	DETAILS
Display	1920 × 1080 minimum; enhanced at 2200px+ and 3000px+
Administrator Rights	Required — installer requests elevation automatically
Network	LAN access to inverter fleet (Gateway) or secure path to gateway (Remote)
Disk Space	Sufficient for database, archives, exports, and forecast model bundles
Ports	3500 (Express API), 9000 (Python FastAPI / Modbus proxy)

Getting Started

Installation

1. Run the installer (`Inverter-Dashboard-Setup-x.x.x.exe`).
2. Accept the Windows UAC prompt to allow administrator elevation.
3. Choose the installation directory (default is recommended).
4. Complete the wizard and launch the application.

First Sign-In

FIELD	DEFAULT VALUE
Username	`admin`
Password	`1234`

> **TIP**
>
> Change your credentials immediately after first login via the **Update Username and Password** link on the sign-in screen. The Authorization Key is required.

Credential Management

ACTION	DESCRIPTION
Update Username and Password	Change credentials using the Authorization Key
Recover with Authorization Key	Restore default credentials (`admin` / `1234`)
Remember credentials	Persists login on the workstation

After successful sign-in, the application maximizes to full screen and begins service initialization. Wait for the startup loading screen to complete before evaluating live values.

Interface Layout

Header Bar

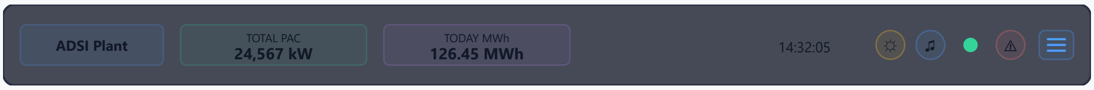

Fig 2 — Header bar layout with global status indicators

ELEMENT	DESCRIPTION
Plant Logo / Title	Identifies the plant and dashboard instance
`TOTAL PAC`	Combined AC power from all online inverters (kW)
`TODAY MWh`	Today's cumulative plant energy (MWh)
Clock & Date	Local workstation time, updates every second
Theme Toggle	Cycle between Dark, Light, and Classic themes
Alarm Sound	Mute or unmute alarm audio notifications
Connection Indicator	Green = live streaming; Red = disconnected
Alarm Notification Hub	Bottom-right summary pill (per-severity tally + count of unacknowledged active alarms); click to open the quick-ACK panel. The sidebar `ALARMS` item carries the matching count badge for navigation.
Menu Button	Toggle the side navigation panel

Side Navigation

PAGE	PURPOSE
Inverters	Live status, node-level control, real-time output
Analytics	Daily energy with interval charts and day-ahead comparison
Forecast	Day-ahead generation and Solcast management
Alarms	Episode history with severity, duration, ACK

PAGE	PURPOSE
Energy	5-minute interval energy by inverter
Audit	Operator command history
Report	Daily performance with availability
Export	Export to Excel or CSV
Settings	Config, connectivity, license, updates, backup

Visual Themes

Dark (default) Light Classic

Inverters Page

The primary live operations workspace for monitoring the full inverter fleet, reviewing node-level status, and sending control commands.

Toolbar Controls

CONTROL	FUNCTION
Inverter Filter	All, Online, Offline, or Alarmed
Layout	Auto, 2–7 column grid
Show/Hide Cap	Toggle the plant output cap panel

Fleet Status Chips

CHIP	MEANING
Inverters	Active / Total
Nodes	Active / Total
Online	Currently producing
Alarmed	Active unACK'd alarms
Offline	Not responding

PAC Output Band Legend

≥90% High   >70% Moderate   >40% Mild   ≤40% Low   Alarm (blink)

Inverter Card

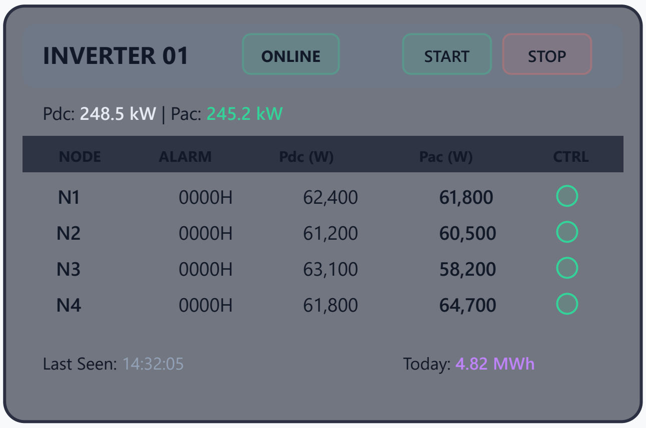

Fig 3 — Inverter card with node-level telemetry and controls

Node Controls

ACTION	SCOPE	AUTHORIZATION
Node button	Single node	Confirmation only
Card Start/Stop	Entire inverter	Confirmation only
Bulk START/STOP	Selected set	Authorization key required

Bulk Inverter Commands

FIELD	FUNCTION
Inverter Ranges	Enter `1-13, 16, 18, 23-27`
All Inverters	Auto-fill full range
Clear	Clear selection
START / STOP SELECTED	Execute on selected inverters

IMPORTANT

Bulk control requires a time-sensitive authorization key. Isolated inverters are skipped automatically.

Plant Output Auto Capping

Automatically regulates total plant output within a defined MW band by sequentially stopping and starting whole inverters.

Cap Configuration

FIELD	FUNCTION
Upper Limit (MW)	Threshold that triggers automatic stop
Lower Limit (MW)	Threshold for restarting stopped inverters
Sequence	Ascending, Descending, or Exemption mode
Exempted Inverters	Comma-separated exclusion list
Cooldown (s)	Settling time between actions (default: 30s)

Cap Actions

ACTION	FUNCTION
Preview Plan	Simulates next decision using live state
Enable Cap	Arms the controller (requires authorization)
Disable Monitoring	Stops automation without restarting stopped inverters
Release Controlled	Restarts stopped inverters sequentially and ends session

Cap Status Panel

FIELD	DESCRIPTION
Status	Active, Idle, or Cooldown
Reason	Human-readable trigger explanation
Last Action	Timestamp of most recent event
Cooldown	Remaining time before re-engagement
Curtailed	Total MW removed
Controllable	Inverters available for dispatch
Pending	In-flight actions
Exempted	Excluded numbers

IMPORTANT

- Whole-inverter sequential control only
- Narrow bands may not settle cleanly
- Non-exempted inverters stay under controller authority while cap is enabled
- Manual control is blocked for non-exempted inverters until cap is disabled
- In Remote mode, cap actions proxy to the gateway

Analytics Page

Interval-based production review with day-ahead forecast comparison and weather context.

CONTROL	FUNCTION
Date	Select the day to analyze
Interval	5 min, 15 min, 30 min, 1 hour
Load View	Load data for selected date and interval

Summary Metrics

METRIC	MEANING
Total Energy	Plant total (MWh)
Peak Output	Highest AC power (kW)
Reporting Inverters	Count with data
Latest Interval	Last recorded timestamp

Selected Date Summary

ITEM	MEANING
Actual MWh	Authoritative total daily energy
Day-ahead MWh	Forecasted total
Variance MWh	Difference between actual and forecast
Peak Interval	Highest interval energy with timestamp

> **LIVE UPDATES**
>
> When today is selected, charts and summary auto-refresh with each server push. Past dates load on demand.

Day-Ahead Generator

Gateway only. Enter **Days** and click **Generate**. Blocked in Remote mode.

Automatic Day-Ahead Schedule

The system automatically generates tomorrow's day-ahead forecast on a fixed cron schedule. Each run checks the existing forecast's **quality** before deciding whether to regenerate.

TIME	ROLE
04:30	Early morning — first pass with overnight Solcast data
09:30	Pre-cutoff — catches weather data refreshes before the 10:00 AM control room submission deadline
18:30	Post-solar-day — refreshes with full day of actual generation data
20:00	Evening re-check
22:00	Final nightly pass

Quality Gate

Before each cron run, the system classifies the existing forecast into one of these quality states:

QUALITY	MEANING	ACTION
healthy	Complete forecast, correct provider, fresh Solcast input	Skip — no regeneration
missing	No forecast rows exist	Generate
incomplete	Fewer slots than the solar window requires	Regenerate
wrong_provider	Generated with a different provider than currently configured	Regenerate
stale_input	Solcast data has been refreshed since the forecast was built	Regenerate
weak_quality	Last run failed or variant is unknown	Regenerate

SOLCAST FRESHNESS DETECTION

The quality gate compares the Solcast snapshot timestamp used at generation time against the current snapshot timestamp. If Solcast has published updated weather data since the last generation (e.g., weather changes after the 04:30 run), the 09:30 cron will detect the stale input and regenerate automatically before the 10:00 AM cutoff.

ACTUALS ISOLATION

The quality assessment only reads forecast predictions, audit metadata, and Solcast weather snapshots. It never reads today's running energy actuals (energy_5min, inverter readings, or intraday adjustments). Live production data cannot interfere with the quality gate or trigger false regenerations.

7-Day Weather Outlook

Daily cards: sky condition, temperature range, rainfall, cloud %, wind speed, solar irradiance. Serves cached data when the API is unavailable.

Day-Ahead vs Reality — Locked @ Previous 10 AM

Multi-series comparison chart showing the frozen 10 AM day-ahead forecast snapshot (P10/P50/P90 confidence band) against intraday Solcast updates, plant actual output, and ML final forecast as the day unfolds. Displays spread % metrics, variance vs P50, and % of actual output within the confidence band. Useful for validating forecast accuracy and detecting late-day weather shifts.

Forecast Page

Dedicated workspace for forecast configuration, Solcast validation, and toolkit preview.

Forecast Sources

SOURCE	USE
Local ML	LightGBM-trained forecasting from historical data
Solcast	External forecast for validation or standalone use

Solcast Access Modes

MODE	USE
Toolkit Login	Chart feed via account sign-in
API Key	Formal API credentials and resource ID

Solcast Settings

FIELD	DESCRIPTION
Plant Resource ID	Unique Solcast site identifier
Forecast Days	1–7 (default: 2)
Resolution	PT5M, PT10M, PT15M, PT30M, PT60M
Toolkit Email / Password	Account credentials

Forecast Tuning

Manual overrides for ML engine training parameters. Leave blank to use the engine's own defaults.

FIELD	PURPOSE	RANGE	DEFAULT
Est-actual Weight	Override the satellite est-actual training weight (how much weight the engine gives to Solcast estimated-actual values when training the model). Validate new values with a backtest before relying.	0.50–1.00	Auto (engine-tuned)
Intraday Blend Max	Cap how strongly intraday observed vs. day-ahead corrections are blended (0 = no intraday blending, 1 = maximum). Validate with a backtest.	0.00–1.00	0.72

ML TRAINING ENGINE — DAY-AHEAD GENERATION

The day-ahead forecast uses a **trained LightGBM model** (default since v2.4.40) that learns your plant's generation profile from historical data. It runs on the gateway and produces next-day forecasts from weather inputs and observed patterns.

How Training Works

- Collects historical irradiance, weather, and production data
- Applies an **operational constraint mask** to exclude abnormal periods
- Rejects outliers, flat-line artifacts, and low-confidence samples
- Produces full-day forecasts at configurable intervals

Plant Cap Awareness

- Cap-dispatched stops (scope `plant-cap`) identified separately from manual stops
- During cap-only periods, engine **reconstructs** full-capacity output using physics baseline
- Preserves high-irradiance training samples; only manually constrained slots are excluded

Intraday Adjustment & Error Memory

- Compares observed vs forecast to compute an adjustment ratio (cap-depressed observations excluded)
- Rolling error correction tracks recent accuracy, excluding all operationally constrained slots

SCENARIO	EFFECT ON TRAINING
Normal operation	All data used
Manual stop / fault	Slots excluded
Plant cap stop	Reconstructed to full capacity
Mixed (cap + manual)	Slots excluded

NOTE

Day-ahead generation is only available on the Gateway workstation.

Forecast Performance Monitor

Real-time diagnostics of ML training health, forecast accuracy trends, and data quality flags (v2.4.42+). Defaults to **collapsed**.

Health Chips

CHIP	MEANING
ML Training	Training state and ML backend (LightGBM/sklearn)
Last Run	Timestamp of last generation run
Provider	Active provider (ML or Solcast)
Recent Quality	Quality classification of latest forecast

Accuracy Charts

CHART	SHOWS
Compare Chart	Forecast vs Actual with confidence band (P10/P90)
WAPE Chart	Daily Weighted Absolute Percentage Error trend

Solcast Tri-Band Integration (v2.5.0+)

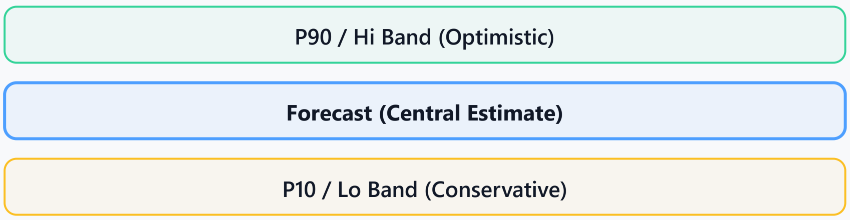

Fig 4 — Solcast tri-band confidence envelope (P10 / Forecast / P90)

Exposes P10/Lo and P90/Hi percentile forecasts, feeding 6 additional LightGBM features:

FEATURE	DESCRIPTION
`solcast_lo_kwh`	P10 energy (conservative)
`solcast_hi_kwh`	P90 energy (optimistic)
`solcast_lo_vs_physics`	P10 vs physics ratio

FEATURE	DESCRIPTION
`solcast_hi_vs_physics`	P90 vs physics ratio
`solcast_spread_pct`	% spread between P10 and P90
`solcast_spread_ratio`	Spread as ratio of central estimate

BACKWARD COMPATIBILITY

Legacy models auto-align with zero-spread fallback. Feature count: 62 → 68. P10/P90 available only from Solcast Toolkit for future dates.

Reliability Dimensions (v2.4.33+)

DIMENSION	KEY	EFFECT
Weather regime	clear/mixed/overcast/rainy	Per-regime bias and reliability
Season	dry/wet	Season-aware blend coefficients
Time-of-day	morning/midday/afternoon	Per-slot blend modulation
Trend	improving/stable/degrading	Blend ±6–8%, damping adjustment

Alarms Page

COLUMN	MEANING
Alarm Time	When alarm was raised
Inverter / Node	Affected equipment
Alarm Code	Hex code from register
Severity	Critical, Warning, Info
Description	Human-readable text
Cleared / Duration	Resolution time and total duration
Status / Ack	Active or Cleared; Acknowledged state

Alarm Episode Model

- New alarm = new episode
- Bitmask change on active node = same episode

- ACK state preserved across changes

Quick-ACK System

- Toast notification (12s TTL) with inline ACK button
- Bottom-right notification hub panel with ACK for up to 50 recent alarms
- Auto-dismiss 1.2s after ACK
- Sound triggers after **5 seconds** unacknowledged
- Mute button silences audio only, not visual alerts

Parameters Page

Per-node 5-minute parameter log for a single inverter at a time. Each tab on the page is one node; rows are 5-minute snapshots of every electrical and operational reading the dashboard captures. Renamed from **Energy** in v2.10.x — the underlying `energy_5min` / `inverter_5min` tables are untouched, while a new `inverter_5min_param` table feeds this view via `dailyAggregator.js`.

Controls

CONTROL	FUNCTION
Inverter	Pick which inverter to view. The page is blank until an inverter is selected.
Date	Today streams live 5-minute samples; past dates load from history.
Mode badge	`LIVE` while today's slots are streaming; `HISTORY` once a past date is loaded.
Solar window	Right-side badge `Solar window: HH:MM–HH:MM` showing which slots are clipped.
Refresh	Re-fetch the selected day without changing the date.

Per-Node Columns (ISM-compatible)

`Pdc` · `Vdc` · `Idc` · `Vac1..3` · `Iac1..3` · `Pac` · `CosΦ` · `Freq` · `parcE` (partial-energy hardware counter snapshot at slot end) · `Alarm` (decoded hex active during the slot) · `Temp` (when register mapping is published).

> **LIVE BEHAVIOR**
>
> For today's date the active node's table appends a new row at every 5-minute boundary; row count ticks up automatically. For past dates the page caches the data until you change the inverter, change the date, or press Refresh.

For a workbook-style export of the same data, see the Daily Data Export card on the Export page.

Audit Page

Operator command accountability log. All columns support sorting with inline filters.

COLUMN	MEANING
Date/Time	Command timestamp
Operator	Recorded identity
Inverter / Node	Affected equipment
Action	START, STOP, etc.
Reason	User request, Cap dispatch, Cooldown, etc.
Scope	`single` , `inverter` , `selected` , or `plant-cap`
Result / IP	OK or ERROR; source workstation address

The `plant-cap` scope badge distinguishes automated dispatch from manual commands.

Report Page

COLUMN	MEANING
Energy (MWh)	Daily total
Peak Pac (kW)	Highest AC output
Avg Pac (kW)	Average during window
Uptime (h)	Operating hours
Alarms	Episode count
Availability (%)	Dynamic window-based uptime ratio
Performance (%)	Actual vs expected output ratio

AVAILABILITY CALCULATION

Per inverter using dynamic window: Start = first non-zero PAC at/after 5 AM; End = last interval at/before 6 PM. `Availability = Online Uptime / Dynamic Window × 100%`

PLANT AVAILABILITY

TOTAL row averages across **all configured inverters**, including inactive ones (0% contribution).

Export Page

Alarm History

Records with code, severity, duration, ACK state.

Energy Summary

Daily energy by inverter and reporting node, with optional `Etotal_kWh` / `parcE_kWh` hardware-counter columns.

Daily Data (v2.10.x)

Per-inverter Excel workbook, one sheet per node, ISM-compatible column order. Today is locked until the End-of-Day snapshot hour.

Day-Ahead Comparison

Forecast vs actual, standard and average table.

Operational Data

Interval-sampled inverter operating data.

Operator Audit

Command history with scope and result.

Daily Performance

Multi-day summary with availability and alarm counts.

- Exports written to configured export folder
- Formats: Excel (.xlsx with styled headers) and CSV
- Cancel button available during active exports
- Result shows filename, size, and timestamp

Energy Summary — Hardware Counter Columns (v2.10.x)

When the operator enables them, the Energy Summary export adds `Etotal_kWh` (lifetime hardware kWh counter at slot end) and `parcE_kWh` (partial-energy counter), plus `Counter_Source` (`eod_clean` / `poll` / `pac_seed`) and quarantine flags. Hardening rules:

- **Today** — delta = current snapshot − today's baseline regardless of source. Partial-day starts line up with PAC-integrated `Total_MWh` on the same row.
- **Today fallback** — if today's baseline row is missing, yesterday's `eod_clean` snapshot is used as the anchor.
- **Past day** — same-day `eod_clean` − baseline preferred; if missing, the next day's open is used as the close-out anchor.
- **Sanity ceiling** — every accepted delta must be ≥ 0 and bounded by 9 000 kWh per unit per day. Outliers blank out and the day total NaN-propagates so the operator notices.

> **PAC REMAINS AUTHORITATIVE**
>
> Hardware counters are reconciliation aids only — they never overwrite the running PAC integration. See `server/hwCounterDeltaCore.js` + `server/tests/hwCounterDeltaCore.test.js` for the locked rules.

Daily Data Export (v2.10.x)

Per-inverter workbook with one sheet per configured node, in ISM-compatible column order so historical data can be cross-referenced with the vendor software.

- **Today is locked** until the dashboard reaches the End-of-Day snapshot hour (HTTP 423 returned otherwise). Until then only past dates may be exported.
- Sheets named `Node 1` , `Node 2` , … built from the IP Configuration `units[invId]` map.
- Slots are clipped to the configured solar window so dawn/dusk noise is excluded.
- Streaming output via `ExcelJS.WorkbookWriter` keeps memory bounded; cancel safely stops the writer mid-stream.
- Filename format matches every other export: `DD-MM-YY Inverter N Daily Data.xlsx` (single day) or `DD-MM-YY-DD-MM-YY Inverter N Daily Data.xlsx` (range).
- The `Temp (°C)` column is **blank by design in v2.10.x** — INGECON SUN does not expose inverter heatsink temperature on the standard FC04 register block. Schema reserves the column so a future register decode populates it without a migration.

Gap Detection (v2.10.x)

Operators can confirm a date's export is complete *before* shipping the workbook by querying:

```
GET /api/params/:inverter/:slave/coverage/:date
```

The response reports expected vs present slots, the missing-slot list as plant-local `HH:MM` ranges, and a `status` of `complete` / `partial` / `empty`. Aggregator drop-sample reasons (`samplesDroppedOffline`, `samplesDroppedStaleTs`, `fieldClampCount`, ...) are also surfaced under `aggregator.*` on `GET /api/system/heartbeat`, so a partial day can be traced back to inverter downtime, clock skew, or out-of-order frames.

Anchor source — HW counter trust ladder (v2.10.x)

The `Counter_Source` column on the Energy Summary export and the `Anchor` pill on Inverter Clocks → Per-Unit Counter Health report which data source today's baseline came from.

PILL	DB SOURCE	MEANING	TRUST
CLEAN	`eod_clean` + today's snapshot captured	Yesterday's clean close anchored today AND today's EOD snapshot has been captured.	★★★★ Best
EOD	`eod_clean`	Yesterday's clean close anchored today; today's EOD snapshot pending after the configured EOD hour.	★★★ Comparable
EOD-ONLY	`eod_clean_only`	Late-created row from a dark-window capture — morning baseline unknown so this day's own Δ is unknown. Used as tomorrow's anchor only. Self-heals to EOD/CLEAN tomorrow.	★ Day Δ unknown
POLL	`poll`	Today's baseline came from the first poll of the day, NOT yesterday's clean close. Etotal Δ undercounts today's energy by whatever the inverter produced before the first poll. PAC-integrated Total stays authoritative.	★★ Δ undercounts
SEED	`pac_seed`	Reserved slot from the v2.9.0 design; no code path currently writes it.	n/a (unused)

SELF-HEALING (V2.10.X)

The dark-window snapshot capture now uses INSERT-or-UPDATE so a fresh-boot gateway no longer loses yesterday's close. Right after a successful eod_clean write, the system re-evaluates today's row: if today is `POLL` and yesterday's eod_clean is now available, today's baseline is rewritten in-place to `eod_clean` and re-anchored to yesterday's true close — the pill flips from POLL to EOD within seconds. Day-total HW columns NaN-propagate for `EOD-ONLY` days so partial-coverage exports never silently report 0 kWh.

Stop Reasons & Serial Number Setting (v2.10.x)

Two new Settings cards expose vendor-FC diagnostics and writes that previously required the ISM laptop. Both surfaces auto-mirror in remote mode but execute their wire actions only on the gateway.

Stop Reasons

Per-node `StopReason` diagnostic captured via vendor FC `0x71` SCOPE peek. The card has two tabs:

- **Captured Snapshots** — pick an inverter, hit **Refresh now** (read-only, no bulk-auth in v2.10.x). The dashboard also **auto-captures** a snapshot on every alarm transition (500 ms staging delay, 30-second per-node cooldown).
- **Lifetime Counters** — cached `ARRAYHISTMOTPARO` histogram of 30 stop motives plus `TOTAL`.

Each captured row links back to its `alarm_id` via the `alarms.stop_reason_id` foreign key, so the alarm drilldown panel can pull the contextual stop reason without a second read.

Serial Number Setting

Mirrors ISM's `frmSetSerial` flow byte-for-byte. Four tabs:

- **Read / Edit / Send** — FC11 *Report Slave ID* read mints a 5-minute session token. *Send* performs UNLOCK (`0xFFFA = 0x0065, 0x07A7`) → WRITE (`0x9C74`) → readback verify. Bulk-auth required, plus the active session token. *Verify Serial Number to send* (default-on) does a fleet-wide FC11 scan and rejects duplicates before the unlock frame.
- **Plant Serial Map** — parallel FC11 sweep across every `(inverter, slave)` with duplicate-detection badges. Honors a 5-minute cache; *Bypass cache* available.
- **Bulk Fix** — re-writes the whole plant against the **locked factory serial map**, generated 1:1 from the authoritative `docs/Fixed_Inverter_SerialNumbers.xlsx` (the permanent field guide). *Show target map* displays all 27 inverters (`T` is the nameplate, reference only; slaves 1–4 are written). *Scan & diff* reads every node and classifies it `match` / `mismatch` / `unreachable` / `no live unit` , with an **Origin** per mismatch (`factory / unknown` or `⇄ Inv X / Node Y` for a physically relocated module). Only mismatches are selectable. *Apply selected* writes each one sequentially with readback verify, logged to `serial_change_log` (scope `bulk`) with timestamp, operator, old→new, and a module-origin note. The per-write uniqueness scan is skipped because the factory map is pre-validated globally unique.
- **Firmware Map** — the nodes **within one inverter** should all run the **same firmware**; each inverter is judged only against *its own* nodes (per-inverter, never plant-wide). A *projection of the serial scan* (firmware rides the same FC11 payload — no extra bus traffic). The compared value is the **inverter firmware code** (`AAV1003xx` , the **Firmware** column) — verified 2026-05-19 by decompiling ISM's own FC11 parser (`IngeconModbusSlaveID_Freescale` : serial + one firmware code, no display-firmware field for this hardware). The **Aux ID 1/2** columns (the `AAS…` strings)

are *unverified auxiliary identifiers* — diagnostics only, **not** compared. *Scan firmware* reads the fleet (bulk-auth, gateway-only); *Show last snapshot* and *Drift log* read the replicated `inverter_firmware_state` / `firmware_drift_log` and work in remote mode. Per-inverter pill: **all same** / **mixed nodes** (a node disagrees with its siblings — post-board-swap signature, dual of the serial relocation guard) / **partial read** / **no read**. Drift is logged only on a real inverter-firmware change of a previously-seen node (Aux-ID-only changes are never drift).

OVERRIDE GATE

Duplicate serials block the write before unlock; an intentional duplicate (e.g. RMA replacement) can be sent with `override_conflicts`. Both Motorola (12 ASCII) and Texas TI (32 ASCII) firmware formats are supported.

READ-BACK IS THE SOURCE OF TRUTH

An identity write makes the inverter re-init its Modbus stack, so the FC16 write *response* is often lost even though the write landed. The pipeline always does the verify read-back (longer settle + retries when the write wasn't ACKed) and judges the result by what the unit actually reports: a confirmed read-back is **success** even with a lost ACK (flagged `write_ack_lost`); only an unchanged read-back is a real failure; if nothing can be read back the result is `write_unconfirmed` (never "success") — rescan to confirm. This eliminates the false `write_failed` where a serial had actually been written.

ONE CREDENTIAL, END TO END

The whole Serial Number feature — single Read / Send, Read-all, Plant Serial Map, Bulk Fix plan/apply, and the duplicate override — uses **one** key: the authorization key (`adsiMM`) — the same rolling key used for every privileged action across the dashboard. Pure read surfaces (audit log, migration history, cached map, target map) need no key.

LOCKED NUMBERING & MODULE RELOCATION

The xlsx is the single source of truth — fix it there and regenerate if a field serial is wrong. Every node's serial is fixed even when a physical node is absent (the serial stays reserved, never reused), which is what lets the dashboard recognise a **physically moved module** by its serial. When you transfer a power module (e.g. Inverter 4 / Node 1 → Inverter 27 / Node 2), Bulk Fix auto-detects it, requires the auto-shown *Acknowledge relocated module(s)* tick, and records the board's origin slot in the change log with correct timestamps for historical traceability.

MEASUREMENT BOARD (POWER MODULE) MIGRATION HISTORY

The *Load migration history* button in the Bulk Fix tab shows the full chronological trail of relocated Measurement Boards — origin slot → where the board was found → the serial written, with timestamp, operator, and result. Structured origin columns (`origin_inverter` / `origin_node`) back the query, and even a detected-but-not-yet-acknowledged relocation is recorded so a board's movement is never lost. It captures relocations re-serialized via either Bulk Fix or the one-by-one Send tab. *Export* (with an Excel/CSV selector) saves the whole trail through the standard export pipeline — styled `.xlsx` or `.csv` in the usual Logs tree (`All Inverters\Audits\`) with the same date-aware filename as every other export, opening the folder when done. Read-only and available in remote mode (proxied + fetched locally just like the Alarms/Audit exports).

PER-INVERTER ONLY & FIELD CHECK

Comparison is strictly **per inverter** — a node is "different" only when its inverter firmware code disagrees with the other nodes on the **same** inverter; there is no plant-wide firmware reference. An unreadable node never overwrites its stored snapshot. The Utility / Calibration Tool exposes a single-inverter *Firmware Check* for the connected unit (slaves 1–4), so a technician can confirm a freshly swapped board matches its siblings without a full plant scan.

Firmware Upgrade (Experimental)

IRREVERSIBLE — READ FIRST

Flashing rewrites the inverter DSP program and **cannot be undone**. A wrong or interrupted image can brick the unit. This feature is gated, experimental, and available **only in the standalone Utility Tool** (never the dashboard). Use it only with the manufacturer-supplied image for the exact unit, after operator sign-off.

A **FW Upgrade** button at the top-right of the per-node controls row (standalone calibrator only) opens the **Firmware Upgrade** dialog, presented as a guided **4-step wizard** — *File → Target → Dry-run → Flash*. A progress rail shows each step's status (done / current / locked) and a later step stays locked until its prerequisite is met, so the irreversible step cannot be reached by accident:

1. **Connect a transport** to the target inverter — **Ethernet** (Modbus-TCP via the transparent gateway) or **RS485-USB** (Modbus-RTU, the most-direct link to the node DSP). The flash uses whichever is connected; a serial flash takes exclusive ownership of the COM port for the duration (reconnect afterwards). The `0x96` baud-bump frame is never sent, so there is no baud-switch race.

Bus-lock note: for an Ethernet flash the dashboard's live poller is a second Modbus master on the same gateway. The flash publishes a cross-process claim so the dashboard **automatically pauses polling of that one inverter** for the duration (shown as *maintenance*, not offline, and not counted as downtime) and resumes the moment the flash ends. This prevents the two-master collision that otherwise makes the DSP reject the start with *"firmware load start (0x90) error code 2"*. The claim self-expires (~2 min) if the calibrator crashes, so polling can never be permanently silenced.
Transport status: the dialog shows a transport-status line near the top — **green** when a link is open, **amber** with an inline **Connect...** shortcut when none is — so you know *before* clicking whether **Read Identity** and **FLASH (live)** can reach the unit; both guide you to connect rather than failing with a cryptic error. (Dry-Run is hardware-free and needs no transport.)

2. **Browse... to the firmware file** via the native OS picker (or paste an absolute path). It must be a `.S` Motorola S-record whose filename follows the `LLLnnnn…` rule; the server still verifies it is a real `.S` file, size-capped, and SHA-256-pinned to the dry-run.
3. **Set the target node** (1–247; broadcast/0 forbidden) and optionally **Read Identity** — an FC11 *Report Slave ID* showing the exact serial / model / running firmware of the unit you would flash. When a file is already chosen, the readout also states whether the flash is an **upgrade**, **downgrade**, or **no change** (unit version → file version), mirroring ISM's pre-flash advisory so you confirm the direction before arming.
4. **Dry-Run (safe).** Simulates the entire flash against an in-memory DSP with **zero hardware contact**, displays the file's **SHA-256**, and verifies that the firmware **code embedded in the image matches the filename** (ISM's `VerificaFicheroFirmware` *"Invalid firmware"* guard), so a renamed or corrupt file is rejected here. A successful dry-run of the exact selected file/node/parameters is a hard precondition for arming the live flash; changing any of them re-disables it.
5. **Arm.** The live block appears only after a passing dry-run. You must tick the irreversible-acknowledgement box and supply the authorization key (`adsiMM`).
6. **FLASH (live).** The flash runs as a background job with a live progress bar, an audit-event log, and an **Abort** button.

Server-side gates

Every gate is enforced server-side regardless of the UI. The live flash is refused unless *all* hold: explicit irreversible confirmation; a prior successful dry-run of the same SHA-256; exactly one link (a TCP host OR a serial COM port, never both/neither); a single non-broadcast node; a verified file (real regular file, `.S`, size-capped, ISM filename rule, SHA-256 match); the **embedded firmware-code match** (the code stored in the image must match the filename — ISM `VerificaFicheroFirmware`); an FC11 model/version compatibility check (apparent downgrades blocked unless **Allow downgrade** is ticked — downgrade *is* possible, see note below); an exclusive RS-485 bus lock; an audit sink; and a watchdog deadline.

BOOTLOADER PRESERVATION & FAIL-SAFE ABORT

The loader never transmits the bootloader or reset-vector banks, so an interrupted or aborted application flash leaves the unit **re-flashable**. **Abort** is fail-safe and intentionally *not* behind an auth prompt so it is always immediately available. While a flash runs the dialog **cannot be closed** (X, backdrop, and Escape are blocked) so an in-progress irreversible operation is never hidden. After a successful flash the tool re-reads FC11 identity (ISM `Identifica`) and reports the firmware the unit now runs — read-only and best-effort, it never marks a successful flash as failed. Every attempt and outcome is appended to `firmware-audit.jsonl` under `%PROGRAMDATA%\InverterDashboard\` (`firmware.pre_flash.direction` , `firmware.live.start` / `.ok` / `.fail` , and `firmware.live.verify_ok` / `.verify_warn`).

Downgrade

Downgrading is possible. The DSP bootloader and the on-wire protocol enforce *no* version monotonicity — the loader erases and writes whatever compatible image you send, older or newer. In ISM the upgrade-vs-downgrade decision lives entirely in its application layer (`QueHableAhoraOCalleParaSiempre(newCode, forceDowngrade)` / `CheckCanUpgradeFirmware`), bypassable with ISM's own *force* flag. This tool mirrors that exactly: the downgrade block compares the unit's **authoritative `model_code` version** (the `AAV1003xx` firmware FC11 actually reports) against the file's version trailer, lifted by ticking **Allow downgrade** under Advanced. Once ticked, an older image of the same product family will flash.

DOWNGRADE CAVEAT

The comparison uses the authoritative `model_code` version, *not* the FC11 `AAS…` auxiliary IDs (which are not the running firmware). It does not change what the hardware accepts — the bootloader takes any compatible image. The file↔model compatibility prefix check (`AAV1003` ...), the embedded firmware-code match, and the SHA-pinned dry-run still apply, and bootloader-bank preservation means even a mistaken-direction flash stays recoverable.

Camera Viewer

Live IP camera monitoring integrated into the main dashboard as a draggable card. Supports three streaming modes via a bundled **go2rtc** process manager and a direct FFmpeg transcoding fallback. Available since **v2.6.4**.

Stream Modes

MODE	BACKEND	PROTOCOL	DESCRIPTION
HLS (default)	go2rtc	HTTP Live Streaming	Best compatibility. go2rtc converts RTSP to HLS segments. Latency ~2–5 s. Uses hls.js in browser.
WebRTC	go2rtc	WebRTC peer-to-peer	Ultra-low latency (<1 s). Requires STUN/TURN for NAT traversal. go2rtc acts as SFU.
FFmpeg	Server	MPEG1/TS over WebSocket	Direct RTSP transcode on the Express server. Requires FFmpeg installed on the host. Uses jsmpeg decoder in browser.

CHOOSING A MODE

Use **HLS** for reliable, low-effort streaming. Choose **WebRTC** when sub-second latency matters. Fall back to **FFmpeg** only when go2rtc is unavailable or the camera does not support standard RTSP.

Camera Card

The camera card is a draggable element that participates in the inverter grid layout. It renders in all layout modes (2-col through 5-col grids) and matches the sizing rules of inverter cards.

ELEMENT	LOCATION	FUNCTION
Video viewport	Card body	Live stream display (fills card)
Camera name label	Top-left overlay	Shows camera name (e.g., “Tapo C110 – Live”)
Live indicator	Top-right overlay	Blinking red dot when stream is active
Settings gear	Bottom bar	Opens the Camera Settings modal
Mute/Unmute	Bottom bar	Toggle audio (if available)
Fullscreen	Bottom bar	Toggle fullscreen video view
Loading spinner	Center overlay	Shown while stream is buffering
Error overlay	Center overlay	Displayed with “Retry” button on stream failure

AUTO-RECONNECT

When a stream drops, the player automatically retries every **5 seconds** until the connection is restored.

Camera Settings Modal

A page-level modal opened by clicking the **gear icon** on the camera card. It consolidates all camera and go2rtc configuration in one place.

1. Stream Mode Selector

Three visual mode cards at the top of the modal. Selecting a mode dynamically shows or hides the relevant input sections below.

MODE CARD	SHOWS	HIDES
HLS / WebRTC	go2rtc Connection, go2rtc Service	RTSP Connection
FFmpeg	RTSP Connection	go2rtc Connection, go2rtc Service

2. go2rtc Connection (HLS / WebRTC modes)

FIELD	DEFAULT	DESCRIPTION
Tailscale / Server IP	`100.93.11.9`	IP address of the machine running go2rtc
API Port	`1984`	go2rtc HTTP API port
Stream Key	`tapo_cam`	Stream name configured in `go2rtc.yaml`

3. RTSP Connection (FFmpeg mode)

FIELD	DEFAULT	DESCRIPTION
Camera IP	`192.168.4.211`	IP address of the IP camera
RTSP Port	`554`	Camera RTSP port
Stream Path	`stream1`	Dropdown: `stream1` (High Quality) or `stream2` (Low Quality)
Username	`Adsicamera`	RTSP authentication username
Password	*(empty)*	RTSP authentication password (show/hide toggle)

> **FFMPEG REQUIRED**
>
> FFmpeg mode requires FFmpeg to be installed on the server host and available on PATH. A warning banner is shown inside the modal when this mode is selected.

4. go2rtc Service (HLS / WebRTC modes, Gateway only)

Manage the bundled go2rtc background process directly from the modal. This section is **hidden in Remote mode**.

ELEMENT	DESCRIPTION
Status grid	4-column display: **Status** (stopped/starting/running/error), **PID**, **Crashes** (count), **Health** (last check time)
Auto-start checkbox	When checked, go2rtc starts automatically on server boot. Saved as server setting `go2rtcAutoStart`.
Start button	Starts go2rtc process. Disabled while already running.
Stop button	Gracefully stops go2rtc (SIGTERM, then SIGKILL after 3 s). Disabled while stopped.
Status message	Feedback area for start/stop results and errors.

> **STATUS POLLING**
>
> While the modal is open, the dashboard polls `/api/streaming/go2rtc-status` every **5 seconds** and updates the status grid in real time. Polling stops when the modal closes.

5. Modal Actions

BUTTON	FUNCTION
Reset Defaults	Restores all fields to factory defaults (see defaults table below)
Apply & Connect	Saves settings to localStorage, persists auto-start to server, closes modal, and reconnects the stream

Default Configuration

SETTING	DEFAULT VALUE	STORAGE KEY
Stream Mode	`hls`	`cam_mode`
go2rtc IP	`100.93.11.9`	`cam_go2rtc_ip`
go2rtc API Port	`1984`	`cam_go2rtc_port`
Stream Key	`tapo_cam`	`cam_stream_key`
Camera IP	`192.168.4.211`	`cam_ip`
RTSP Port	`554`	`cam_rtsp_port`
Stream Path	`stream1`	`cam_stream_path`
Username	`Adsicamera`	`cam_user`
Password	*(empty)*	`cam_pass`

go2rtc Service Defaults

SETTING	DEFAULT	NOTES
Auto-start on boot	Disabled (`0`)	Server setting `go2rtcAutoStart`
API bind address	`127.0.0.1:1984`	Localhost only (security)
WebRTC bind address	`127.0.0.1:8555`	Localhost only (security)
Max crash restarts	3	After 3 crashes, status becomes “error”
Health check interval	5 seconds	HTTP GET to go2rtc API
Health check timeout	2 seconds	Per health-check request

go2rtc Process Lifecycle

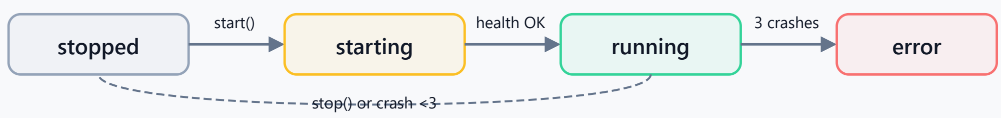

Fig 5a — go2rtc process state machine

Security

- **Localhost-only binding** — go2rtc API (`127.0.0.1:1984`) and WebRTC (`127.0.0.1:8555`) are not accessible from external networks.
- **Gateway-mode only** — go2rtc start is blocked with HTTP 403 in Remote mode. The service section is hidden in the UI.
- **STUN server** — Uses Google’s public STUN server (`stun:stun.l.google.com:19302`) for WebRTC ICE negotiation.

Troubleshooting

ISSUE	CAUSE	RESOLUTION
“Stream failed” error	go2rtc not running or wrong IP/port	Open camera settings, verify go2rtc IP and port, click Start in service section
go2rtc won’t start	Port 1984 or 8555 in use	Stop conflicting process or edit `go2rtc.yaml` ports
WebRTC no video	NAT/firewall blocking	Ensure STUN server is reachable; consider configuring TURN
FFmpeg mode no video	FFmpeg not installed on server	Install FFmpeg and ensure it is on PATH

ISSUE	CAUSE	RESOLUTION
Service section hidden	Remote mode active	Switch to Gateway mode in Settings → Connectivity
Auto-restart stopped	3+ consecutive crashes	Check go2rtc logs, fix RTSP source, then manually restart

Settings

Administrative center with 8 configuration sections: Plant, Data & Polling, Cap Defaults, Connectivity & Gateway Link, Forecast, License, App Updates, and Cloud Backup.

Plant Configuration

FIELD	DEFAULT	USE
Plant Name	ADSI Plant	Display naming
Operator Name	OPERATOR	Audit identity
Inverter Count	27	Total configured
Nodes/Inverter	4	Reporting nodes
Latitude / Longitude	—	Weather context

Connectivity & Gateway Link

FIELD	USE
Operation Mode	Gateway or Remote
Gateway URL	URL of gateway workstation
API Token	Shared token for remote access
Tailscale Hint	Optional device identifier

The **Gateway Link** tab groups live status into **Connection** (mode, gateway URL, gateway reachable, Tailscale, live link, last contact) and **Standby Refresh** (last refresh, rows received, last DB pull, background job), with a Transfer Monitor that shows live RX/TX rates and standby-refresh progress.

Runtime Health

Real-time metrics: CPU %, Memory RSS, Uptime, Poll Ticks, Duration, Fetch Errors, Persist Rows, WS Clients/Drops.

License

FIELD	PURPOSE
Status / Expiry	Current state and expiry date
Upload Replacement	Apply a new license file
Audit Trail	Timestamp, Event, Details history

App Updates

ACTION	FUNCTION
Check for Updates	Query update channel
Download Update	Download package
Restart & Install	Orderly shutdown + installer handoff

IMPORTANT

Do not power off during restart/install handoff. Wait for first poll after restart.

Operating Modes

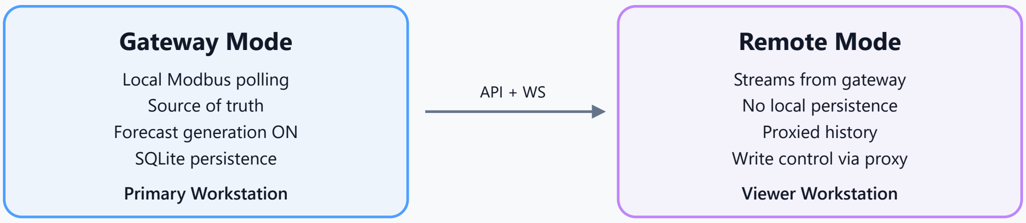

Fig 5 — Gateway vs Remote operating modes

Gateway Link Health States

STATE	MEANING
Connected	Live data streaming normally

STATE	MEANING
Degraded	Elevated latency
Stale	Last snapshot shown
Disconnected	No active connection
Auth Error	Token mismatch or expired
Config Error	Invalid URL or configuration

Connectivity Options

- **Tailscale** — Zero-config mesh VPN (recommended)
- **Traditional VPN** — OpenVPN, WireGuard, IPsec
- **Port Forwarding** — Direct exposure (requires HTTPS + strong tokens)
- **Cloud Relay** — Cloudflare Tunnel, ngrok
- **LAN** — Direct IP on shared network

Operator Messaging

- Floating message bubble in bottom corner
- Stored on gateway, synced to remote workstations
- Audio notification for inbound messages
- Read state tracking per workstation
- Rate limited: 10 per 60s per machine; 500-row retention

Cloud Backup & Restore

PROVIDER	AUTHORIZATION
OneDrive	Microsoft OAuth via Azure Client ID
Google Drive	Google OAuth
S3-compatible	Static access key pair

SETTING	OPTIONS
Schedule	Manual, Daily 3 AM, Every 6 hours
Scope	App data, Config files, Logs (optional)

App data = SQLite DB + forecast model bundles + Solcast artifacts + weather cache.

- Restore creates safety backup first
- Overwrites active database and config
- Restart required to complete restore

Standard Operating Workflows

Daily Startup Check

1. Launch and sign in.
2. Wait for startup loading screen to finish.
3. If Remote and gateway unreachable, choose Gateway or retry in the Connection Mode picker.
4. Confirm license area is clear.
5. Check connection dot, clock, TOTAL PAC, and TODAY MWh.
6. Open Inverters page and confirm fleet status chips.

Live Control

1. Open Inverters page.
2. Select target inverter or node.
3. Review alarm state and data freshness.
4. Send START or STOP action.
5. Confirm via status updates and audit log.

Plant Output Cap

1. Show the Cap panel on Inverters page.
2. Set Upper/Lower Limit, Sequence, Cooldown, and Exemptions.
3. Click Preview Plan to review.
4. Click Enable Cap (authorization required).
5. Monitor Cap Status and Controlled Inverters Table.
6. Release: click Release Controlled. Disarm: click Disable Monitoring.

Daily Performance Review

1. Open Analytics for interval review and day-ahead comparison.
2. Open Energy for interval detail.
3. Open Report for formal summary.
4. Export from Report or Export page.

Remote Standby Refresh

1. Confirm Remote mode.
2. Open Settings → Connectivity & Gateway Link.
3. Click Refresh Standby DB.
4. Allow preflight, then wait for snapshot transfer.
5. Restart to apply staged database.

Cloud Backup

1. Configure and authorize provider.
2. Save settings and select scope.
3. Run Backup Now or rely on schedule.
4. Review backup history.

Camera Setup

1. On the Dashboard, locate the camera card (or drag it into position).
2. Click the gear icon on the camera card to open Camera Settings.
3. Select a stream mode: **HLS** (recommended), **WebRTC**, or **FFmpeg**.
4. For HLS/WebRTC: enter the go2rtc IP, port, and stream key. In the Service section, click **Start** to launch go2rtc.
5. For FFmpeg: enter camera IP, RTSP port, stream path, username, and password.
6. Optionally check **Auto-start on server boot** to persist go2rtc across restarts.
7. Click **Apply & Connect**. The stream should appear in the camera card.
8. If the stream fails, check the troubleshooting table in the Camera Viewer section.

IP Configuration

Open via **Settings → IP Configuration** or Ctrl + I. Auth code required, 1-hour session.

Available in **both Gateway and Remote mode**. On a Remote viewer a banner notes that edits save to the gateway (authoritative) and mirror back locally, so addressing can be updated without remoting into the gateway PC. The local reachability scan and **Check Status** counter are suppressed in Remote mode (those inverters sit on the gateway LAN); the gateway keeps polling them.

COLUMN	DESCRIPTION
Inverter	Label
IP Address	Modbus TCP target
Polling Interval	Refresh rate (ms)
Enabled Units	Active nodes to poll
Loss %	Cable/connection loss (used by forecast engine)

Additional: **Check Status** (test config), **Open Topology** (separate auth, 10-min session), **Theme toggle**.

Keyboard Shortcuts & Tips

SHORTCUT	FUNCTION
Ctrl + T	Topology window
Ctrl + I	IP Configuration
Ctrl + =	Zoom in
Ctrl + -	Zoom out
Ctrl + 0	Reset zoom
Ctrl + L	Theme toggle (aux windows)
Ctrl + R	Force full data reload

- **Alarm notifications**: sidebar `ALARMS` badge shows the count (navigation); the bottom-right hub pill opens the quick-ACK panel
- **Sidebar**: collapses to icons on narrow windows; hover for tooltips
- **Inverter cards**: click to expand detail panel
- **Message bubble**: bottom corner, opens operator messaging

Troubleshooting

SYMPTOM	LIKELY CAUSE	ACTION
Connection dot disconnected	Live link unavailable	Check mode, URL, token, Tailscale
Stale status	Retained snapshot	Check gateway health
Standby DB unchanged	Staged, not active	Restart the application
Newer-local warning	Local copy newer	Cancel or Force Pull
TODAY MWh old after startup	Polling not caught up	Refresh Standby DB, restart, wait
Connection Mode picker	Remote startup failed	Choose Gateway or retry Remote
Day-ahead unavailable	Remote mode	Generate from gateway
Cap "Cannot POST"	Old gateway or wrong URL	Update gateway, verify URL
Cap limits too close	Step > deadband	Increase Upper/Lower gap
Cloud restore unavailable	Provider not ready	Refresh list, verify auth
Topology blocked	Remote mode (IP Config now works remotely; Topology stays Gateway-only)	Use gateway workstation for Topology
IP Config change not applied	Gateway unreachable or rejected the save	Verify gateway URL/token online; the save banner shows the gateway error
Silent alarms	Muted or no audio	Re-enable sound, check audio
Export fails	Path/date/format issue	Verify folder, filters, mode
High WAPE	Stale Solcast or retraining needed	Check Performance Monitor
Camera “Stream failed”	go2rtc not running or wrong IP	Open Camera Settings, verify IP/port, Start service

SYMPTOM	LIKELY CAUSE	ACTION
go2rtc won't start	Port 1984/8555 in use	Stop conflicting process or change ports
Camera service hidden	Remote mode active	Switch to Gateway mode
Inverter nodes stop polling but the IP still pings	The gateway's Modbus session to that inverter has wedged (stale socket / comm-board TCP queue), which a ping cannot detect	Wait for auto-recovery, or click the **Reconnect** button on that inverter's row in **IP Configuration**
"Open device web page" does nothing from a remote viewer	The comm board lives on the gateway LAN and is not reachable from the remote PC	Use the gear icon in **IP Configuration** — it now routes the device page through the gateway automatically

Inverter polling stalls while the IP still pings (v2.11.x)

An inverter can keep answering `ping` while its Modbus polling fails — ping only proves the comm board's network chip is alive, not that the Modbus session behind it is healthy. On the Ingeteam **AAX0041** Ethernet→RS-485 comm board, the board accepts the TCP connection and queues requests, so a stale or wedged session can leave the dashboard's per-inverter connection stuck even though the IP is reachable. Historically the only field fix was to change the inverter's IP, which forced the gateway to build a brand-new connection.

AUTOMATIC RECOVERY

The gateway now watches each inverter's connection. After ~30 seconds with no successful read, it automatically discards the wedged Modbus client and builds a fresh one (rate-limited to once per minute per inverter) — no operator action and no IP change needed. This makes the old "change the IP" workaround unnecessary in normal operation.

MANUAL RECONNECT

To force it immediately, open **IP Configuration** and click the circular **Reconnect** button on that inverter's row (next to Save). This drops and rebuilds only that inverter's gateway socket — it changes no inverter setting and no IP. It works from a remote viewer too (the request is routed to the gateway). If the inverter recovers after a manual reconnect but never on its own, the fault is likely upstream of the gateway (a duplicate IP, or a stale ARP / connection-tracking entry on the network switch/router) rather than the comm board itself.

OPEN DEVICE WEB PAGE (GATEWAY & REMOTE)

The gear icon on each **IP Configuration** row opens the comm board's own web page. On the gateway it opens directly on the plant LAN; from a **remote viewer** it is routed through the gateway automatically (the device is not reachable from the remote PC). The remote route is byte-for-byte, so the device's HTML, styles, scripts and images all render. Tailscale note: the inverter's LAN IP is *not* reachable directly over Tailscale unless a subnet router is configured — the gateway proxy is what makes the page work remotely without one.

Quick Reference

Defaults & Constants

ITEM	VALUE
Default Credentials	`admin` / `1234`
Plant Name	ADSI Plant
Inverter Rated (Pmax) / Dependable (Pnom)	997.64 kW / 906.92 kW
Nodes per Inverter	4 (249.41 kW Pmax / 226.73 kW Pnom each)
Solar Window	5:00 AM – 6:00 PM
Alarm Sound Threshold	5 seconds sustained
Ports	3500 (Express), 9000 (FastAPI)
ML Features	68 columns (with tri-band)

Page Quick Finder

NEED	BEST PAGE
Live status	Inverters
Interval analysis	Analytics + Energy
Day-ahead comparison	Analytics + Export
Forecast setup	Forecast
Forecast accuracy	Forecast Performance Monitor
Alarm review	Alarms
Action traceability	Audit

NEED	BEST PAGE
Daily summary	Report
File generation	Export
Mode / sync	Settings → Connectivity
License / updates	Settings → License / Updates
Backup	Settings → Cloud Backup
Network config	IP Configuration
Messaging	Floating bubble
Live camera feed	Camera card (Dashboard)
Camera / go2rtc config	Camera Settings modal

Data Units

ITEM	UNIT
PAC / PDC	W or kW
Energy	MWh
Duration	sec / min / hrs
WAPE	%

Status Terms

STATUS	MEANING
Online	Communicating normally
Offline	No live data
Stale	Last snapshot shown
Alarm	Active alarm
Acknowledged	Operator-ACK'd
CAP STOPPED	Stopped by cap controller

Day-Ahead Generation Paths

PATH	TRIGGER	AUDIT
Manual UI	POST /api/forecast/generate	Node
Auto scheduler	Python loop → delegate	Node

PATH	TRIGGER	AUDIT
Python CLI	--generate-date	Node
CLI fallback	Node unreachable	Python
Node cron	04:30/09:30/18:30/20:00/22:00	Node

ADSI Inverter Dashboard v2.11.5

Designed & Developed by Engr. Clariden Montaño REE